

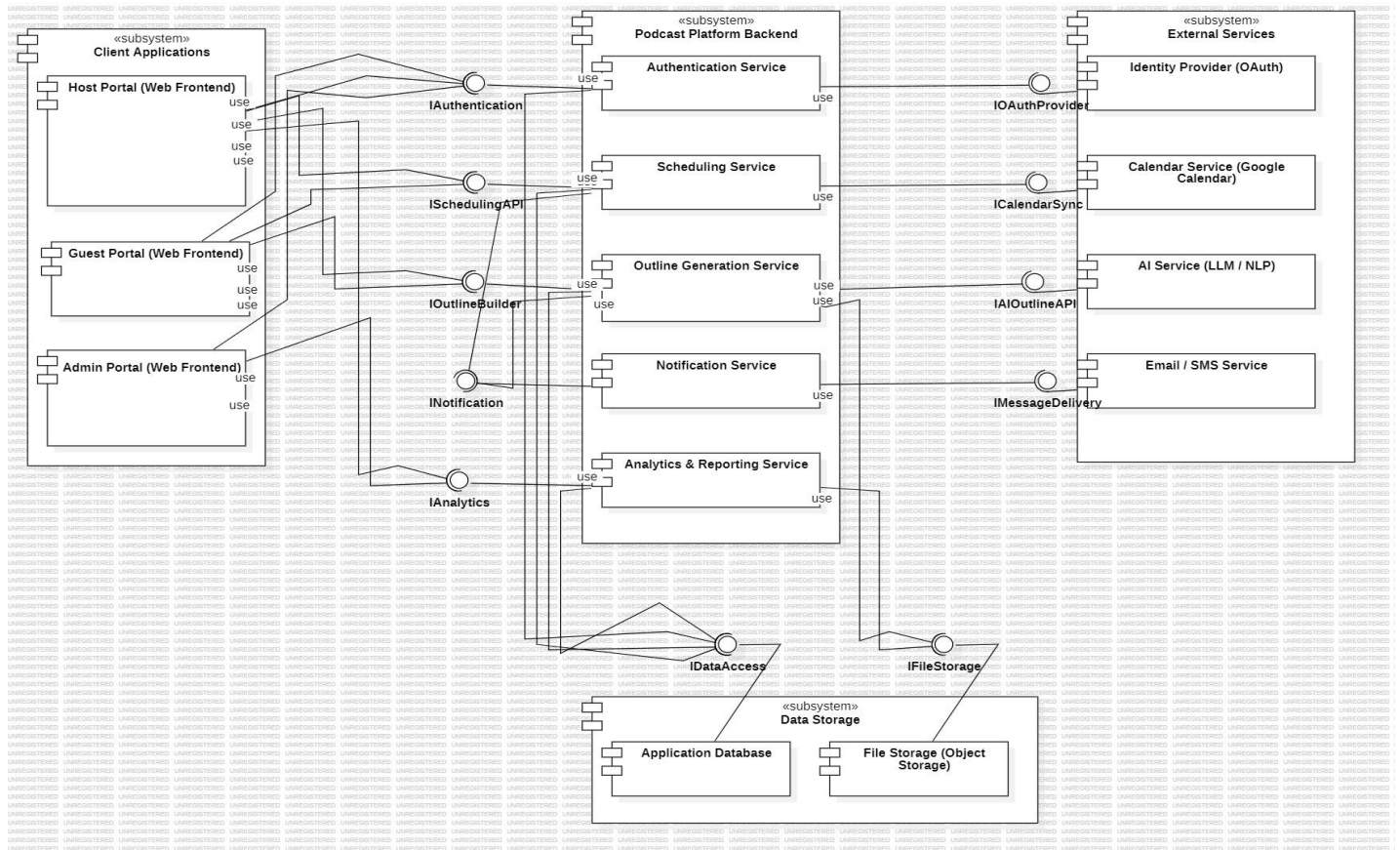
LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Podcast Guest Scheduling & Outline Builder

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UML Component Diagram



ARCHITECTURE SELECTION & JUSTIFICATION

System: Podcast Guest Scheduling & Outline Builder

Architecture Selection

We chose Microservices Architecture for the Podcast Guest Scheduling & Outline Builder System.

Two Scenario-Related Reasons

- 1. Independent scheduling and production services:** Availability, booking, discussion-topic processing, run-of-show generation, and notification functions have distinct responsibilities and can be maintained as separate services.
- 2. Independent scaling and integration:** Scheduling, outline generation, and notifications have different workloads and external integrations such as calendar, AI, and email/SMS services. Separating them permits independent scaling and integration.

Security Advantage

Authentication is isolated as a dedicated service and can control access to scheduling, guest information, discussion topics, biography links, and production notes. This addresses the requirement to protect guest and episode information from unauthorized access.

Performance Benefit

Run-of-show generation can operate as an independent service, allowing resources to be allocated specifically to outline and production-note generation. This supports the requirement for formatted production sheets to be generated in under one second under simulated peak load.